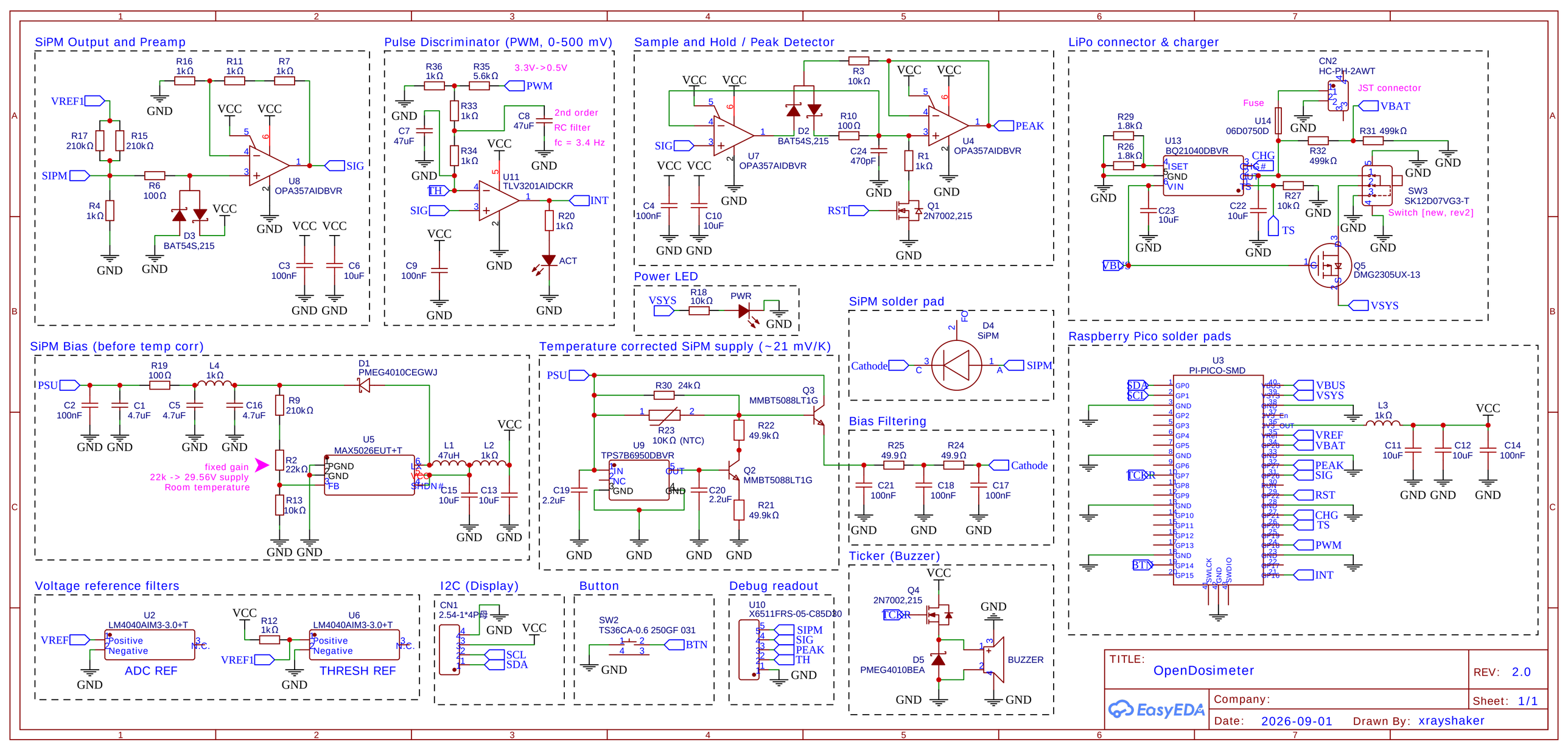

SiPM Output and Preamp
R16 1kΩ
R11 1kΩ
R7 1kΩ
VREF1
GND
VCC
R17 210kΩ
R15 210kΩ
SIPM
R6 100Ω
SIG
U8 OPA357AIDBVR
R4 1kΩ
D3 BAT54S,215
C3 100nF
C6 10uF
Pulse Discriminator (PWM, 0-500 mV)
R36 1kΩ
R35 5.6kΩ
3.3V->0.5V
PWM
R33 1kΩ
C8 47uF
2nd order
RC filter
fc = 3.4 Hz
C7 47uF
R34 1kΩ
U11 TLV3201AIDCKR
TH
INT
R20 1kΩ
ACT
C9 100nF
Sample and Hold / Peak Detector
R3 10kΩ
U7 OPA357AIDBVR
D2 BAT54S,215
R10 100Ω
C24 470pF
R1 1kΩ
U4 OPA357AIDBVR
PEAK
C4 100nF
C10 10uF
RST
Q1 2N7002,215
LiPo connector & charger
CN2 HC-PH-2AWT
JST connector
VBAT
Fuse
U14 06D0750D
R31 499kΩ
R29 1.8kΩ
R26 1.8kΩ
U13 BQ21040DBVR
ISET
VIN
CHG
CHG#
OUT
TS
R32 499kΩ
R27 10kΩ
SW3 SK12D07VG3-T
Switch [new, rev2]
C23 10uF
C22 10uF
VBUS
Q5 DMG2305UX-13
VSYS
Power LED
VSYS
R18 10kΩ
PWR
SiPM solder pad
D4 SiPM
Cathode
SIPM
SiPM Bias (before temp corr)
PSU
R19 100Ω
L4 1kΩ
D1 PMEG4010CEGWJ
C2 100nF
C1 4.7uF
C5 4.7uF
C16 4.7uF
R9 210kΩ
U5 MAX5026EUT+T
fixed gain
22k -> 29.56V supply
Room temperature
R2 22kΩ
R13 10kΩ
L1 47uH
L2 1kΩ
C15 10uF
C13 10uF
Temperature corrected SiPM supply (~21 mV/K)
R30 24kΩ
Q3 MMBT5088LT1G
R23 10KΩ (NTC)
U9 TPS7B6950DBVR
R22 49.9kΩ
Q2 MMBT5088LT1G
R21 49.9kΩ
C19 2.2uF
C20 2.2uF
Bias Filtering
R25 49.9Ω
R24 49.9Ω
C21 100nF
C18 100nF
C17 100nF
Raspberry Pico solder pads
U3 PI-PICO-SMD
SDA
SCL
TCKR
BTN
VBUS
VSYS
VREF
VBAT
PEAK
SIG
RST
CHG
TS
PWM
INT
L3 1kΩ
C11 10uF
C12 10uF
C14 100nF
Voltage reference filters
U2 LM4040AIM3-3.0+T
VREF
ADC REF
R12 1kΩ
U6 LM4040AIM3-3.0+T
VREF1
THRESH REF
I2C (Display)
CN1 2.54-1*4P
SCL
SDA
Button
SW2 TS36CA-0.6 250GF 031
BTN
Debug readout
U10 X6511FRS-05-C85D30
SIPM
SIG
PEAK
TH
Ticker (Buzzer)
Q4 2N7002,215
D5 PMEG4010BEA
BUZZER
TITLE: OpenDosimeter
REV: 2.0
Company:
Sheet: 1/1
Date: 2026-09-01
Drawn By: xrayshaker
EasyEDA